# HCIA-AI V4.0 Certification Practice Exam • Chapter 3

Module 3: Deep Learning & Foundation Models (100 Questions & Answer Key)

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## Question 1

TRUE OR FALSE • 16 Points

Typically, in a deep neural network that uses the Sigmoid activation function, the gradient may decrease close to 0 within five layers, causing severe vanishing gradients.

**A.** True
**B.** False

## Question 2

TRUE OR FALSE • 16 Points

When training a deep neural network with the Softsign function, gradients never saturate regardless of how deep the network is.

**A.** True
**B.** False

## Question 3

TRUE OR FALSE • 16 Points

The Tanh activation function can effectively eliminate and completely solve the vanishing gradient problem in arbitrarily deep neural networks.

**A.** True
**B.** False

## Question 4

TRUE OR FALSE • 16 Points

Batch Normalization is a technique used in deep learning to standardize layer inputs, stabilizing internal covariate shift and accelerating model convergence.

**A.** True
**B.** False

## Question 5

TRUE OR FALSE • 16 Points

A single-layer perceptron without hidden layers is capable of solving linearly non-separable problems such as the XOR logical operation.

**A.** True
**B.** False

**Page Answers:** **Q1**: True • **Q2**: False • **Q3**: False • **Q4**: True • **Q5**: False

### Question 6

TRUE OR FALSE • 16 Points

The maximum derivative of the standard Sigmoid function is 1.0, which occurs when its input x equals 0.

**A.** True

**B.** False

### Question 7

TRUE OR FALSE • 16 Points

The ReLU activation function outputs 0 for all negative inputs ($x < 0$) and has a derivative of 1 for all positive inputs ($x > 0$).

**A.** True

**B.** False

### Question 8

TRUE OR FALSE • 16 Points

During neural network inference (testing), Dropout continues to randomly set 50% of the neurons' outputs to zero.

**A.** True

**B.** False

### Question 9

TRUE OR FALSE • 16 Points

In Convolutional Neural Networks, the pooling layer introduces learnable weight parameters that are updated via backpropagation.

**A.** True

**B.** False

### Question 10

TRUE OR FALSE • 16 Points

Parameter sharing in CNNs means that every single pixel in an image requires a completely unique, independent set of convolution kernel weights.

**A.** True

**B.** False

**Page Answers:** **Q6**: False • **Q7**: True • **Q8**: False • **Q9**: False • **Q10**: False

### Question 11

TRUE OR FALSE • 16 Points

Compared with standard RNNs, CNNs are better suited for extracting 2D spatial grid features from images.

**A.** True
**B.** False

### Question 12

TRUE OR FALSE • 16 Points

Transformer models discard recurrent loops completely and rely on Self-Attention mechanisms and Positional Encoding to capture token dependencies.

**A.** True
**B.** False

### Question 13

TRUE OR FALSE • 16 Points

The Adam optimizer combines the principles of Momentum (first moment estimation) and RMSProp (second moment estimation).

**A.** True
**B.** False

### Question 14

TRUE OR FALSE • 16 Points

LSTM networks contain only a single gate known as the Reset Gate.

**A.** True
**B.** False

### Question 15

TRUE OR FALSE • 16 Points

The Gated Recurrent Unit (GRU) contains five separate gates, which makes it much more computationally complex than standard LSTM.

**A.** True
**B.** False

**Page Answers:** **Q11**: True • **Q12**: True • **Q13**: True • **Q14**: False • **Q15**: False

### Question 16

TRUE OR FALSE • 16 Points

In ResNet, residual shortcut (skip) connections allow gradients to flow directly through identity paths, enabling effective training of networks with hundreds of layers.

**A.** True
**B.** False

### Question 17

TRUE OR FALSE • 16 Points

Exploding gradients during deep network training can lead to numerical overflow and NaN loss values, which can be mitigated by Gradient Clipping.

**A.** True
**B.** False

### Question 18

TRUE OR FALSE • 16 Points

The AdaGrad optimizer monotonically increases the learning rate during training, eventually causing infinite learning rates.

**A.** True
**B.** False

### Question 19

TRUE OR FALSE • 16 Points

In the Transformer architecture, the Scaled Dot-Product Attention divides the dot products of Q and K by the square root of the key dimension (sqrt(d_k)) to prevent softmax gradients from becoming extremely small.

**A.** True
**B.** False

### Question 20

TRUE OR FALSE • 16 Points

BERT is an autoregressive decoder-only language model trained primarily to predict the very next token in a causal sequence.

**A.** True
**B.** False

**Page Answers:** **Q16**: True • **Q17**: True • **Q18**: False • **Q19**: True • **Q20**: False

### Question 21

TRUE OR FALSE • 16 Points

GPT models adopt a bidirectional encoder architecture where each token attends at the same time to all past and future tokens during causal generation.

**A.** True
**B.** False

### Question 22

TRUE OR FALSE • 16 Points

Data augmentation techniques (e.g., random flipping, cropping, and color jittering) reduce training set size and increase model underfitting.

**A.** True
**B.** False

### Question 23

TRUE OR FALSE • 16 Points

Cross-Entropy loss is widely used with the Softmax activation function for multi-class classification neural networks.

**A.** True
**B.** False

### Question 24

TRUE OR FALSE • 16 Points

Backpropagation calculates the gradient of the loss function with respect to each network parameter using the mathematical Chain Rule of calculus.

**A.** True
**B.** False

### Question 25

TRUE OR FALSE • 16 Points

In standard RNNs, Backpropagation Through Time (BPTT) unfolds the recurrent computational graph across time steps to compute weight gradients.

**A.** True
**B.** False

**Page Answers:** **Q21**: False • **Q22**: False • **Q23**: True • **Q24**: True • **Q25**: True

### Question 26

SINGLE-ANSWER QUESTION • 16 Points

During neural network training, which of the following values are continuously updated by optimization algorithms using gradient descent?

- **A.** Sample labels
- **B.** Learning rate
- **C.** Model parameters
- **D.** Batch size

### Question 27

SINGLE-ANSWER QUESTION • 16 Points

The ReLU function is commonly used in deep learning. What is the value range (output range) of the standard ReLU function?

- **A.** (-inf, +inf)
- **B.** (-1.0, +1.0)
- **C.** [0, +inf)
- **D.** (0.0, 1.0)

### Question 28

SINGLE-ANSWER QUESTION • 16 Points

The Sigmoid activation function is monotonic and continuous, but when used in very deep networks, which problem is most likely to emerge?

- **A.** Vanishing gradient
- **B.** Gradient explosion
- **C.** Underfitting caused by high variance
- **D.** Loss function becoming negative

### Question 29

SINGLE-ANSWER QUESTION • 16 Points

Which of the following activation functions can effectively reduce the vanishing gradient problem in positive activation regions?

- **A.** ReLU
- **B.** Tanh
- **C.** Softsign
- **D.** Sigmoid

### Question 30

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT a recurrent neural network model?

- **A.** Long Short-Term Memory
- **B.** Logistic Regression Classifier
- **C.** Gated Recurrent Unit
- **D.** Bidirectional RNN

**Page Answers:** **Q26**: C • **Q27**: C • **Q28**: A • **Q29**: A • **Q30**: B

### Question 31

SINGLE-ANSWER QUESTION • 16 Points

If a 32 x 32 feature map is processed through a 2 x 2 pooling layer with stride = 2 and no padding, what is the size of the output feature map?

- **A.** 30 x 30
- **B.** 32 x 32
- **C.** 64 x 64
- **D.** 16 x 16

### Question 32

SINGLE-ANSWER QUESTION • 16 Points

If a 32 x 32 image is input and a 5 x 5 convolution kernel is applied with stride = 1 and padding = 0, what is the spatial dimension of the output feature map?

- **A.** 27 x 27
- **B.** 30 x 30
- **C.** 28 x 28
- **D.** 32 x 32

### Question 33

SINGLE-ANSWER QUESTION • 16 Points

Which layer in a Convolutional Neural Network achieves translational invariance and reduces spatial dimensions without introducing trainable parameters?

- **A.** Batch Normalization Layer
- **B.** Pooling Layer
- **C.** Convolutional Layer
- **D.** Fully-Connected Layer

### Question 34

SINGLE-ANSWER QUESTION • 16 Points

Which optimizer uses both the exponentially decaying average of past gradients (first moment) and the past squared gradients (second moment)?

- **A.** Standard SGD
- **B.** Adam
- **C.** Momentum SGD
- **D.** AdaGrad

### Question 35

SINGLE-ANSWER QUESTION • 16 Points

In the LSTM cell, which gate is responsible for deciding what percentage of information from the previous cell state (C_{t-1}) should be discarded?

- **A.** Forget Gate
- **B.** Input Gate
- **C.** Reset Gate
- **D.** Output Gate

**Page Answers:** **Q31**: D • **Q32**: C • **Q33**: B • **Q34**: B • **Q35**: A

### Question 36

SINGLE-ANSWER QUESTION • 16 Points

Which gate in the LSTM cell decides which newly computed candidate values should be stored in the updated cell state?

- **A.** Update Gate
- **B.** Forget Gate
- **C.** Output Gate
- **D.** Input Gate

### Question 37

SINGLE-ANSWER QUESTION • 16 Points

What is the primary motivation behind the Residual Connection (shortcut connection) introduced in ResNet?

- **A.** Enabling identity mappings F + x so gradients flow smoothly, preventing network degradation
- **B.** Replacing non-linear activation functions with direct linear identity pass-through layers
- **C.** Flattening two-dimensional spatial feature maps into single-dimension sequential vectors
- **D.** Doubling kernel weight parameter counts per layer to maximize model memorization capacity

### Question 38

SINGLE-ANSWER QUESTION • 16 Points

In the Transformer Self-Attention formula Attention(Q, K, V) = softmax(Q K^T / sqrt(d_k)) V, what does 'd_k' represent?

- **A.** Total number of training samples
- **B.** Total vocabulary size
- **C.** Dimension of the key vector
- **D.** Number of layers in the encoder

### Question 39

SINGLE-ANSWER QUESTION • 16 Points

Why is Positional Encoding essential in the Transformer architecture?

- **A.** To replace dense feedforward projection layers with fixed sinusoidal trigonometric operations
- **B.** To calculate cross-entropy loss gradients across multi-token target vocabulary dictionaries
- **C.** To provide the permutation-invariant self-attention mechanism with sequence order awareness
- **D.** To project sparse token index embeddings into low-dimensional normalized spatial coordinates

### Question 40

SINGLE-ANSWER QUESTION • 16 Points

Which activation function is defined as f(x) = max(0.01x, x) to solve the 'Dying ReLU' problem?

- **A.** Exponential Linear Unit
- **B.** Scaled Exponential Unit
- **C.** Gaussian Error Linear Unit
- **D.** Leaky Rectified Linear Unit

**Page Answers:** **Q36**: D • **Q37**: A • **Q38**: C • **Q39**: C • **Q40**: D

### Question 41

SINGLE-ANSWER QUESTION • 16 Points

What happens when a hidden neuron experiences the 'Dying ReLU' problem?

- **A.** Its weight values explode toward positive infinity, triggering numerical NaN overflow errors
- **B.** Its activation function spontaneously transitions into a continuous hyperbolic tangent curve
- **C.** Its input stays negative, causing 0 output and 0 gradient, permanently halting parameter updates
- **D.** Its internal bias coefficient inverts polarity, forcing the activation to saturate at positive one

### Question 42

SINGLE-ANSWER QUESTION • 16 Points

Which loss function is mathematically defined as L = - sum(y_i * log(p_i)) for multi-class classification?

- **A.** Mean Squared Error
- **B.** Cross-Entropy Loss
- **C.** Hinge Loss
- **D.** Mean Absolute Error

### Question 43

SINGLE-ANSWER QUESTION • 16 Points

In Convolutional Neural Networks, what is a 1x1 convolution commonly used for?

- **A.** Doubling the spatial height and width dimensions of feature maps through bilinear upsampling
- **B.** Cross-channel pooling to reduce or expand channel depth while injecting non-linear computation
- **C.** Extracting directional edge gradients across spatial diagonal pixel coordinates in input images
- **D.** Binarizing continuous feature activations into discrete zero-or-one boolean threshold outputs

### Question 44

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of the Output Gate in an LSTM unit?

- **A.** Discarding obsolete historical context from the previous cell state vector during reset steps
- **B.** Resetting the internal forget gate bias to zero at the start of every sequence
- **C.** Controlling which filtered portion of the updated cell state is exposed as hidden state h_t
- **D.** Scaling the input embedding matrix by the inverse square root of hidden dimension

### Question 45

SINGLE-ANSWER QUESTION • 16 Points

What is the key advantage of Multi-Head Attention over single-head attention in Transformers?

- **A.** It processes multiple independent batch sequences at the same time using a single shared linear matrix
- **B.** It projects inputs into diverse representation subspaces, allowing joint attention to different features
- **C.** It eliminates the requirement for positional encodings by computing temporal ordering dynamically
- **D.** It replaces floating-point matrix multiplications with low-precision integer bitwise operations

**Page Answers:** **Q41**: C • **Q42**: B • **Q43**: B • **Q44**: C • **Q45**: B

### Question 46

SINGLE-ANSWER QUESTION • 16 Points

In deep learning, what does 'Weight Decay' directly correspond to in terms of loss regularization?

- **A.** L1 Regularization
- **B.** Dropout probability
- **C.** L2 Regularization
- **D.** Learning rate scheduling

### Question 47

SINGLE-ANSWER QUESTION • 16 Points

What regularization method temporarily removes units (both hidden and visible) along with their incoming and outgoing edges with probability p during each training iteration?

- **A.** Dropout
- **B.** Gradient clipping
- **C.** Batch normalization
- **D.** Feature scaling

### Question 48

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the training procedure of the BERT foundation model?

- **A.** K-Means clustering of raw pixels
- **B.** Masked Language Modeling and Next Sentence Prediction
- **C.** Supervised reinforcement learning from human feedback
- **D.** Next token prediction only using causal masking

### Question 49

SINGLE-ANSWER QUESTION • 16 Points

Which classical CNN architecture won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) in 2012, sparking the modern deep learning boom?

- **A.** ResNet-152
- **B.** Transformer
- **C.** LeNet-5
- **D.** AlexNet

### Question 50

SINGLE-ANSWER QUESTION • 16 Points

What is the primary function of the 'C' (Cell State) vector in an LSTM network?

- **A.** Acting as an unobstructed conveyor belt that carries linear memory across steps with additive updates
- **B.** Calculating cross-entropy objective loss against ground truth label targets on every recurrent step
- **C.** Applying softmax normalization directly to the unscaled input gate logits at each step
- **D.** Multiplying hidden state vectors by a fixed independent rotation matrix at each step

**Page Answers:** **Q46**: C • **Q47**: A • **Q48**: B • **Q49**: D • **Q50**: A

### Question 51

SINGLE-ANSWER QUESTION • 16 Points

In feedforward neural networks, what mathematical operation connects an input vector x to layer output z before activation?

- **A.** Polynomial square root: z = sqrt
- **B.** Linear affine transformation: z = W x + b
- **C.** String concatenation
- **D.** Division by zero

### Question 52

SINGLE-ANSWER QUESTION • 16 Points

Why are non-linear activation functions mandatory between successive layers in deep neural networks?

- **A.** To initialize all neural weight parameters uniformly to constant values of positive one point zero
- **B.** To convert floating-point tensor activations into discrete binary booleans for hardware caching
- **C.** To prevent multiple stacked linear layers from mathematically collapsing into a single linear map
- **D.** To restrict operating system memory consumption and prevent CUDA kernel deadlocks during training

### Question 53

SINGLE-ANSWER QUESTION • 16 Points

Which optimizer maintains an exponential moving average of past gradients to help accelerate in directions with persistent gradients and dampen oscillations?

- **A.** Nearest Neighbor Search
- **B.** Standard SGD without momentum
- **C.** Exhaustive Grid Search
- **D.** Momentum Optimizer

### Question 54

SINGLE-ANSWER QUESTION • 16 Points

In an image with 3 color channels (RGB) of size 224 x 224, what are the dimensions of a single convolution kernel of spatial size 3 x 3?

- **A.** 1 x 1 x 3
- **B.** 3 x 3 x 3
- **C.** 224 x 224 x 3
- **D.** 3 x 3 x 1

### Question 55

SINGLE-ANSWER QUESTION • 16 Points

What is the formula for calculating output dimension O given input size W, kernel size K, padding P, and stride S?

- **A.** O = floor - 1
- **B.** O = floor
- **C.** O = floor + 2
- **D.** O = floor + 1

**Page Answers:** **Q51**: B • **Q52**: C • **Q53**: D • **Q54**: B • **Q55**: D

### Question 56

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is a key architectural characteristic of the Swish activation function f(x) = x * sigmoid(beta * x)?

- **A.** It restricts all activated output values strictly within the closed real interval of zero and one
- **B.** It eliminates backward gradient propagation by computing forward parameter updates analytically
- **C.** It truncates all non-zero floating-point numbers into eight-bit signed integer representations
- **D.** It is smooth and non-monotonic, consistently reducing gradient vanishing in deep architectures

### Question 57

SINGLE-ANSWER QUESTION • 16 Points

Which layer normalization technique in deep learning normalizes activations across features for each individual sample, widely used in Transformer models?

- **A.** Batch Normalization
- **B.** Min-Max Normalization
- **C.** Layer Normalization
- **D.** Weight Normalization

### Question 58

SINGLE-ANSWER QUESTION • 16 Points

What is 'Early Stopping' in neural network training?

- **A.** Halting model training when validation performance ceases improving, thereby preventing overfitting
- **B.** Deleting intermediate training checkpoints automatically whenever the loss curve plateaus on disk
- **C.** Terminating model optimization immediately after epoch one regardless of error convergence rates
- **D.** Continuing gradient updates until training loss reaches zero across all epochs

### Question 59

SINGLE-ANSWER QUESTION • 16 Points

Which architecture is specifically designed for sequence-to-sequence tasks (e.g., machine translation) using two separate RNN or Transformer blocks?

- **A.** Single Decision Tree
- **B.** Encoder-Decoder Architecture
- **C.** Linear Hyperplane Sorter
- **D.** K-Means Clusterer

### Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

In the Transformer architecture, which mechanisms are integrated into each sub-layer to stabilize gradient flow and accelerate deep network training? (Multiple Choice)

- **A.** Residual connection
- **B.** Max pooling
- **C.** Layer normalization
- **D.** Convolution kernel

**Page Answers:** **Q56**: D • **Q57**: C • **Q58**: A • **Q59**: B • **Q60**: AC

### Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following factors contribute to the Vanishing or Exploding Gradient problem during backpropagation through time (BPTT) in standard RNNs?

- **A.** Repeated matrix multiplications of weight matrices across extended sequential time steps
- **B.** Use of saturating activation functions like Sigmoid whose derivatives are bounded
- **C.** Operating on fixed-width non-overlapping spatial convolutional sliding windows
- **D.** Long-range temporal dependencies that exponentially attenuate or escalate backpropagated error

### Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent recognized methods of Parameter-Efficient Fine-Tuning (PEFT) for large foundation models?

- **A.** Full Model Parameter Retraining on entire multi-terabyte web crawl datasets
- **B.** LoRA injecting low-rank decomposition matrices
- **C.** Prefix Tuning and Prompt Tuning conditioning model activations with learnable vectors
- **D.** Re-initializing all transformer feedforward projection layers from random uniform distributions

### Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding LoRA (Low-Rank Adaptation) fine-tuning are correct?

- **A.** It freezes the original pre-trained model weight matrices during fine-tuning
- **B.** It completely removes the multi-head attention mechanism from Transformer layers
- **C.** It introduces trainable low-rank decomposition matrices alongside attention projections
- **D.** It greatly reduces the number of trainable parameters and GPU memory consumption

### Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following activation functions have strictly bounded output ranges that squash inputs into finite numerical intervals?

- **A.** Sigmoid function
- **B.** Tanh function
- **C.** Standard ReLU function
- **D.** Leaky ReLU function

### Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding deep neural network capacity, parameters, and generalization are TRUE?

- **A.** Increasing parameter capacity generally enhances the network's ability to fit complex non-linear functions
- **B.** Excessive parameter capacity relative to training sample size increases the risk of overfitting
- **C.** Regularization techniques help high-capacity models generalize better
- **D.** Data augmentation artificially expands training diversity to help prevent overparameterized memorization

**Page Answers:** **Q61**: ABD • **Q62**: BC • **Q63**: ACD • **Q64**: AB • **Q65**: ABCD

### Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following activation functions are symmetric, bounded S-shaped curves commonly used in neural networks?

- **A.** Sigmoid function
- **B.** Hyperbolic Tangent or Tanh function
- **C.** Standard Rectified Linear Unit
- **D.** Leaky ReLU function

### Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

In the Transformer scaled dot-product attention formula, which vectors are multiplied together via dot-product to compute raw compatibility scores?

- **A.** Query vector
- **B.** Value vector
- **C.** Key vector
- **D.** Residual shortcut vector

### Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following algorithms are commonly used first-order gradient-based optimizers in deep learning?

- **A.** RMSProp
- **B.** Momentum SGD
- **C.** K-Means
- **D.** Adam

### Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following components are foundational architectural building blocks of Convolutional Neural Networks (CNNs)?

- **A.** Recurrent hidden-to-hidden sequential transition matrices
- **B.** Convolutional layers with learnable spatial filter kernels
- **C.** Pooling layers
- **D.** Non-linear activation functions

### Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized regularization techniques used to prevent overfitting in deep neural networks?

- **A.** Early stopping based on validation loss monitoring
- **B.** L1 and L2 weight decay regularization
- **C.** Data augmentation expanding training sample diversity
- **D.** Dropout randomly deactivating neurons during training

**Page Answers:** **Q66**: AB • **Q67**: AC • **Q68**: ABD • **Q69**: BCD • **Q70**: ABCD

### Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding the Adam optimizer are correct?

- **A.** It tracks an exponentially decaying average of past gradients
- **B.** It completely eliminates the necessity of error backpropagation gradient evaluation
- **C.** It tracks an exponentially decaying average of past squared gradients
- **D.** It includes bias-correction terms to counteract zero-bias initialization

### Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are the primary pooling operations commonly used in Convolutional Neural Networks?

- **A.** Max Pooling
- **B.** Median Value Sorting Filter
- **C.** Average Pooling
- **D.** Cross-Entropy Loss Normalization

### Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Long Short-Term Memory (LSTM) are true?

- **A.** LSTM introduces dedicated gate mechanisms to reduce vanishing gradients in standard RNNs
- **B.** The Cell State acts as a linear conveyor belt across long time sequences
- **C.** The Forget Gate controls information retention from previous cell state
- **D.** LSTM can only be used for image classification and cannot process text

### Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are key architectural advantages of the Transformer over classical RNN models?

- **A.** Ability to scale efficiently to billions of parameters
- **B.** Direct global dependency modeling without distance-related attenuation
- **C.** Eliminating the need for any training data
- **D.** Full parallelization during training across all sequence tokens

### Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are classic deep Convolutional Neural Network architectures?

- **A.** VGGNet
- **B.** LeNet-5
- **C.** AlexNet
- **D.** ResNet

**Page Answers:** **Q71**: ACD • **Q72**: AC • **Q73**: ABC • **Q74**: ABD • **Q75**: ABCD

### Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following structural mechanisms are integrated inside a standard Transformer Encoder block?

- **A.** Multi-Head Self-Attention layer
- **B.** Depthwise spatial 2D max-pooling subsampling
- **C.** Layer Normalization and residual shortcut connections
- **D.** Position-wise Feedforward Neural Network

### Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized mathematical limitations of the classic Sigmoid activation function?

- **A.** Vanishing gradient problem when inputs have large absolute values
- **B.** Outputs are not zero-centered, causing zigzag gradient updates during backpropagation
- **C.** Output values unbounded toward positive infinity
- **D.** Complete absence of first-order mathematical differentiability

### Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are typical foundation model methods?

- **A.** Autoregressive language models
- **B.** Single-perceptron linear classifiers
- **C.** Encoder-decoder sequence models
- **D.** Masked autoencoder language models

### Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Dropout are correct?

- **A.** During evaluation/testing, all neurons are kept active and outputs are scaled appropriately
- **B.** It randomly sets activations of selected neurons to zero with probability p during training
- **C.** It prevents complex co-adaptation of feature detectors
- **D.** It is an unsupervised clustering algorithm

### Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are standard loss functions used for multi-class classification and continuous regression respectively?

- **A.** Triplet Contrastive Distance Metric
- **B.** Silhouette Cluster Separation Index
- **C.** Categorical Cross-Entropy Loss
- **D.** Mean Squared Error

**Page Answers:** **Q76**: ACD • **Q77**: AB • **Q78**: ACD • **Q79**: ABC • **Q80**: CD

### Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Batch Normalization are true?

- **A.** It introduces learnable scale and shift parameters
- **B.** It normalizes each feature dimension across the mini-batch during training
- **C.** It allows higher learning rates and accelerates training convergence
- **D.** It can only be used on the final output classification layer

### Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Backpropagation (BP) are correct?

- **A.** It generates random training labels for supervised datasets
- **B.** It computes parameter gradients backward from the output layer to the input layer
- **C.** It relies on the mathematical chain rule of differentiation
- **D.** Gradients computed by BP are passed to optimizers to update model weights

### Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements correctly describe the standard ReLU activation function $f(x) = \max(0, x)$?

- **A.** Its output is strictly bounded within the closed real range of negative one and positive one
- **B.** It possesses a smooth, non-linear curve with continuous second-order derivatives everywhere
- **C.** For positive inputs, its derivative is constant 1, preventing gradient vanishing
- **D.** For negative inputs, its output is constant 0, inducing sparse network representation

### Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are classic Parameter-Efficient Fine-Tuning (PEFT) methods for adapting large pre-trained models?

- **A.** LoRA
- **B.** Prompt Tuning
- **C.** Full retraining
- **D.** Random initialization

### Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following problems can occur when training very deep neural networks without appropriate normalization or skip connections?

- **A.** Faster convergence
- **B.** Vanishing gradients
- **C.** Network degradation
- **D.** Exploding gradients

**Page Answers:** **Q81**: ABC • **Q82**: BCD • **Q83**: CD • **Q84**: AB • **Q85**: BCD

### Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about the Softmax function are correct?

- **A.** Each output probability is strictly bounded between 0 and 1
- **B.** It converts real-valued logits into a probability distribution
- **C.** It can only be used when there are exactly two classes
- **D.** The sum of all output probabilities across classes equals exactly 1

### Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Unlike LSTM which contains three gates, the Gated Recurrent Unit (GRU) simplifies recurrent gating into which distinct gates?

- **A.** Forget Gate
- **B.** Reset Gate
- **C.** Output Gate
- **D.** Update Gate

### Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are benefits of Data Augmentation in Computer Vision?

- **A.** Helps prevent model overfitting
- **B.** Deletes corrupted image files from cloud storage
- **C.** Increases the diversity of training samples without gathering new data
- **D.** Improves the generalization ability of neural networks

### Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common types of Positional Encodings used to provide sequence order awareness in Transformers?

- **A.** Sinusoidal encodings
- **B.** Learned Positional Embeddings
- **C.** Stochastic Gaussian Noise Additive Perturbations
- **D.** Batch Normalization Running Variance Scaling

### Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are hyperparameters of a Convolutional Layer?

- **A.** Padding size
- **B.** Kernel size
- **C.** Stride
- **D.** Number of output channels

**Page Answers:** **Q86**: ABD • **Q87**: BD • **Q88**: ACD • **Q89**: AB • **Q90**: ABCD

### Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary structural layers commonly found in Convolutional Neural Networks (CNNs)?

**A.** Convolutional Layers
**B.** Pooling Layers
**C.** Fully Connected Layers
**D.** Recurrent Feedback Memory Units

### Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Backpropagation and Gradient Descent in deep learning are TRUE?

**A.** Backpropagation applies the calculus Chain Rule to compute partial derivatives of loss with respect to weights
**B.** Error gradients are propagated backward from the output layer through hidden layers toward the input
**C.** Weight parameters are updated in the opposite direction of the gradient to minimize the loss function
**D.** The gradient vector points mathematically in the direction of steepest ascent of the loss surface

### Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, all user-defined neural network models must implement their forward computational graph logic inside the def ____(self, x): method.

**Answer:** _____

### Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch autograd, reverse-mode automatic differentiation is executed by calling loss.____() on the computed scalar loss tensor.

**Answer:** _____

### Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, to prevent gradient accumulation across successive training iterations, developers must call optimizer.____() before loss.backward().

**Answer:** _____

**Page Answers:** **Q91**: ABC • **Q92**: ABCD • **Q93**: forward • **Q94**: backward • **Q95**: zero_grad

### Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the optimizer method called to update learnable model parameters according to computed gradients is optimizer.____().

**Answer:** _____

### Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, to deactivate the autograd gradient computation engine and conserve memory during model inference, code is wrapped inside with torch.____():.

**Answer:** _____

### Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, switching the neural network into evaluation mode (deactivating Dropout and freezing Batch Normalization running statistics) is done by calling model.____().

**Answer:** _____

### Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the standard loss function in torch.nn used for multi-class classification that implicitly combines LogSoftmax and NLLLoss is nn.____().

**Answer:** _____

### Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, to save learned model weights to disk, developers extract the parameter dictionary using model.____().

**Answer:** _____

**Page Answers:** **Q96**: step • **Q97**: no_grad • **Q98**: eval • **Q99**: CrossEntropyLoss • **Q100**: state_dict